



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

AA 2025/2026

**Esistenza di cammini in cui
sono presenti archi rossi e archi blu
(variante di Floyd-Warshall)**

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[Problema]

INPUT:

- $G = (V, E)$, grafo orientato, dove $V = \{1, 2, \dots, n\}$
- $Col : E \rightarrow C$, con $C = \{\text{"red"}, \text{"blue"}, \text{"green"}, \text{"black"}\}$

OUTPUT:

per ogni coppia di vertici i e j , verificare se esiste un cammino P_{ij} in cui sono presenti archi rossi e archi blu

[Sottoproblemi per i vertici i e j]

Sottoproblema di dimensione (k, r, b)

Verificare se esiste un cammino $P_{ij}^{k,r,b}$ da i a j in cui gli archi rossi sono presenti ($r=1$) o assenti ($r=0$) e gli archi blu sono presenti ($b=1$) o assenti ($b=0$), tale che se $k > 0$ eventuali vertici intermedi appartengono a $\{1, 2, \dots, k\}$ e se $k = 0$ non ci sono vertici intermedi

$0 \leq k \leq n, \quad r, b \in \{0, 1\}$

Numero di sottoproblemi per i e j: $4(n+1)$

Numero totale di sottoproblemi: $4(n+1)n^2$

[Coefficienti per i vertici i e j]

Coefficiente del sottoproblema di dimensione (k, r, b)

$e_{ij}^{k,r,b}$ è **TRUE** se e solo se esiste un cammino $P_{ij}^{k,r,b}$ da i a j in cui gli archi rossi sono presenti ($r=1$) o assenti ($r=0$) e gli archi blu sono presenti ($b=1$) o assenti ($b=0$), tale che se $k > 0$ eventuali vertici intermedi appartengono a $\{1, 2, \dots, k\}$ e se $k = 0$ non ci sono vertici intermedi

$0 \leq k \leq n, \quad r, b \in \{0, 1\}$

Numero di coefficienti per i e j: $4(n+1)$

Numero totale di coefficienti: $4(n+1)n^2$

[Coefficienti per i vertici i e j]

Coefficiente del sottoproblema di dimensione (k, r, b)

$e_{ij}^{k,r,b}$ è **TRUE** se e solo se esiste un cammino $P_{ij}^{k,r,b}$ da i a j in cui gli archi rossi sono presenti ($r=1$) o assenti ($r=0$) e gli archi blu sono presenti ($b=1$) o assenti ($b=0$), tale che se $k > 0$ eventuali vertici intermedi appartengono a $\{1, 2, \dots, k\}$ e se $k = 0$ non ci sono vertici intermedi

$0 \leq k \leq n, \quad r, b \in \{0, 1\}$

Numero di coefficienti per i e j: $4(n+1)$

Numero totale di coefficienti: $4(n+1)n^2$

Risultato per i e j $\equiv e_{ij}^{n,1,1}$

[Eq. Ricorrenza - Casi base]

no **rossi**, no **blu**

Sottoproblemi di dimensione ($k = 0, r=0, b=0$)

$$i = j \Rightarrow e_{ij}^{k,r,b} = \text{TRUE}$$

$$i \neq j \wedge (i,j) \in E$$

$$\text{Col}(i,j) \neq \text{"red"} \wedge \text{Col}(i,j) \neq \text{"blue"} \Rightarrow e_{ij}^{k,r,b} = \text{TRUE}$$

$$\text{Col}(i,j) = \text{"red"} \vee \text{Col}(i,j) = \text{"blue"} \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

[Eq. Ricorrenza - Casi base]

yes **rossi**, no **blu**

Sottoproblemi di dimensione ($k = 0, r=1, b=0$)

$$i = j \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \in E$$

Col(i,j)="red"

$$\Rightarrow e_{ij}^{k,r,b} = \text{TRUE}$$

Col(i,j)≠"red"

$$\Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

[Eq. Ricorrenza - Casi base]

Sottoproblemi di dimensione ($k = 0, r=1, b=0$)

$$i = j \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \in E$$

Col(i,j)="red"

$$\Rightarrow e_{ij}^{k,r,b} = \text{TRUE}$$

Col(i,j)≠"red"

$$\Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

[Eq. Ricorrenza - Casi base]

no **rossi**, yes **blu**

Sottoproblemi di dimensione ($k = 0, r=0, b=1$)

$$i = j \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \in E$$

Col(i,j)="blue"

$$\Rightarrow e_{ij}^{k,r,b} = \text{TRUE}$$

Col(i,j)≠"blue"

$$\Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

[Eq. Ricorrenza - Casi base]

yes **rossi**, yes **blu**

Sottoproblemi di dimensione ($k = 0, r=1, b=1$)

$$i = j \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \in E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

$$i \neq j \wedge (i,j) \notin E \Rightarrow e_{ij}^{k,r,b} = \text{FALSE}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0, r, b \in \{0,1\}$



Vertici intermedi in $\{1,2,\dots,k-1, k\}$

$$e_{ij}^{k,r,b} =$$

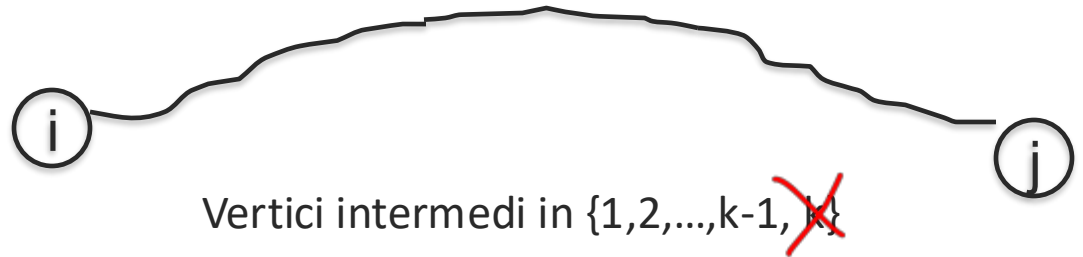
$$\Rightarrow \exists P_{ij}^{k,r,b}$$

Eq. Ricorrenza – Passo ricorsivo

$$k > 0, r, b \in \{0, 1\}$$

$$k \notin P_{ij}^{k,r,b}$$

$$\Rightarrow e_{ij}^{k,r,b} =$$



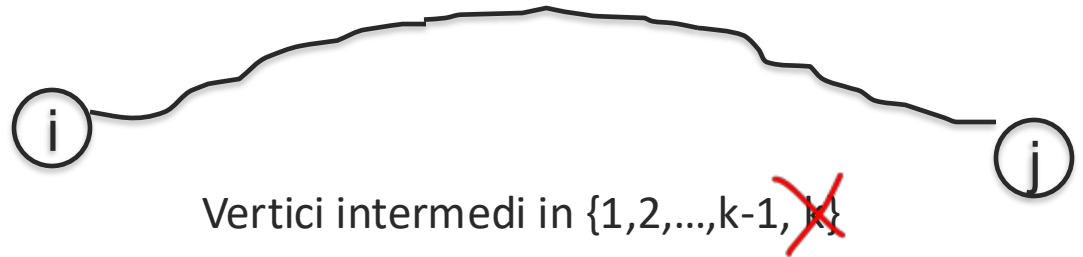
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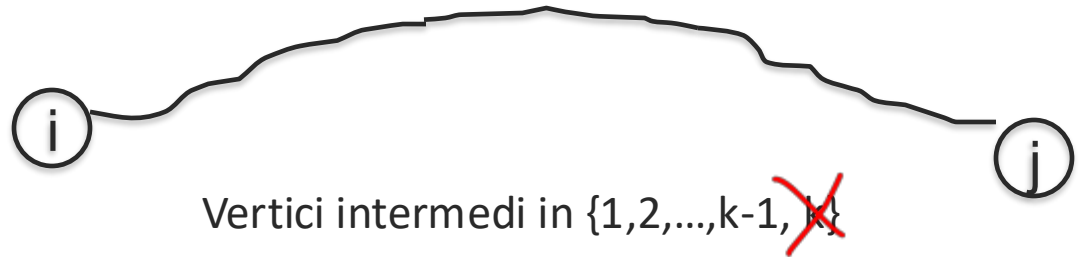
$$\exists P_{ij}^{k-1,r,b} \Rightarrow \exists P_{ij}^{k,r,b}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

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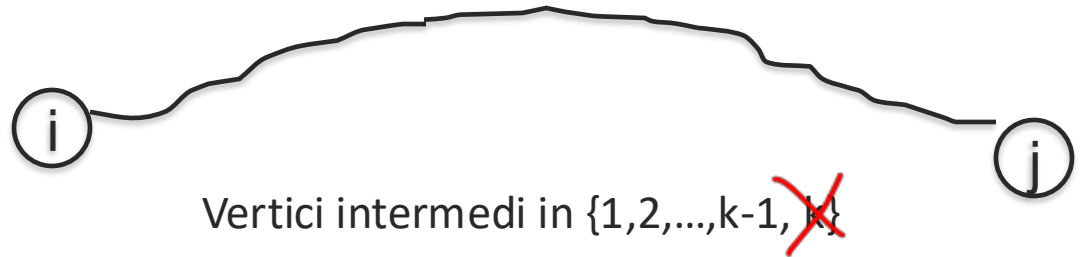
$e_{ij}^{k-1,r,b} \Rightarrow \exists P_{ij}^{k,r,b}$

Eq. Ricorrenza – Passo ricorsivo

$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$\Rightarrow e_{ij}^{k,r,b} =$



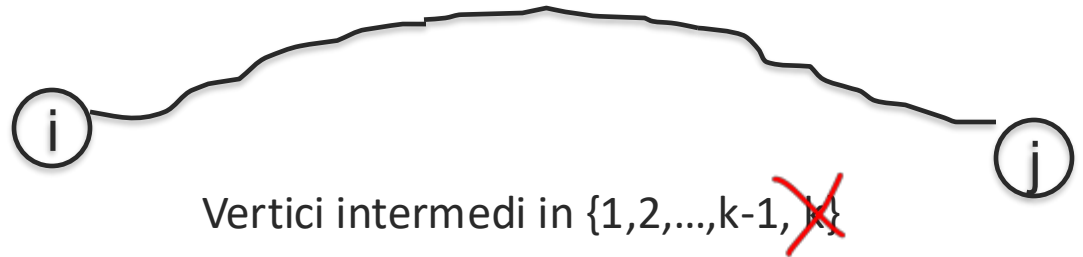
$$e_{ij}^{k-1,r,b} \Rightarrow e_{ij}^{k,r,b}$$

Eq. Ricorrenza – Passo ricorsivo

$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$



$e_{ij}^{k-1,r,b} \Rightarrow e_{ij}^{k,r,b}$

Eq. Ricorrenza – Passo ricorsivo

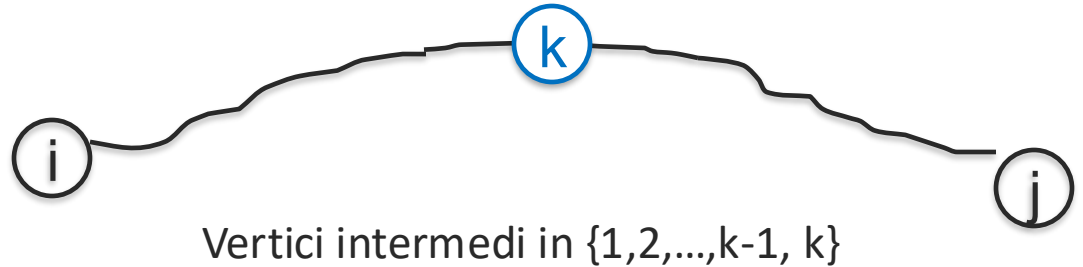
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Eq. Ricorrenza – Passo ricorsivo

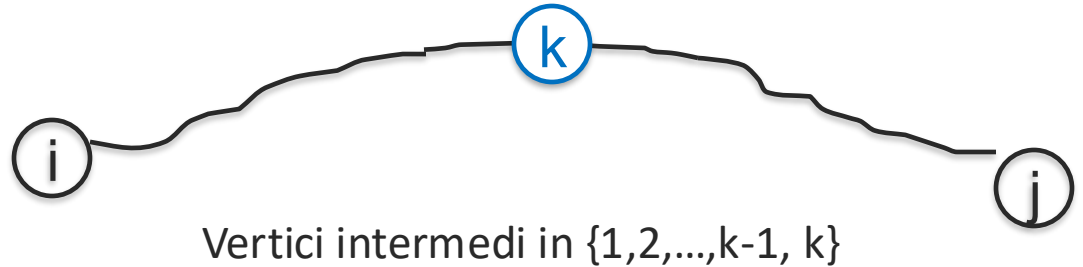
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$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=0, b=0$$

$$\Rightarrow e_{ij}^{k,0,0} =$$



$$\Rightarrow \exists P_{ij}^{k,0,0}$$

Eq. Ricorrenza – Passo ricorsivo

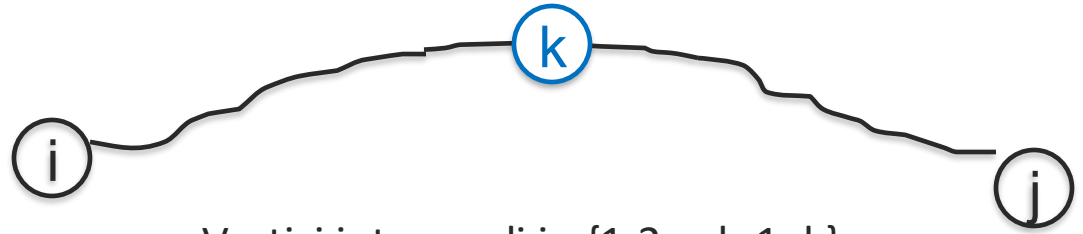
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$

$k \in P_{ij}^{k,r,b}, r=0, b=0$

$\Rightarrow e_{ij}^{k,0,0} =$



$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,0,0}$$

Eq. Ricorrenza – Passo ricorsivo

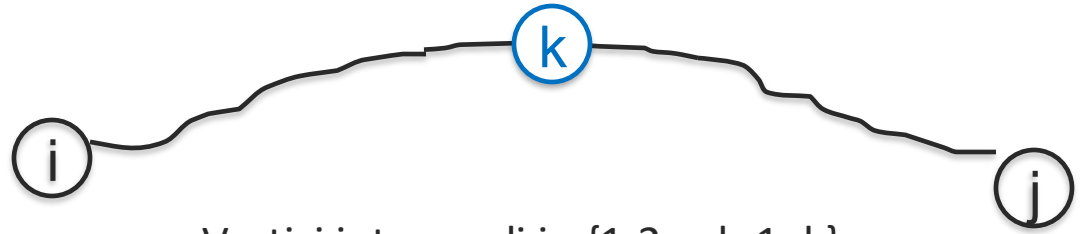
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$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=0, b=0$$

$$\Rightarrow e_{ij}^{k,0,0} =$$



$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,0,0}$$

$$(f = 0 \wedge f' = 0) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

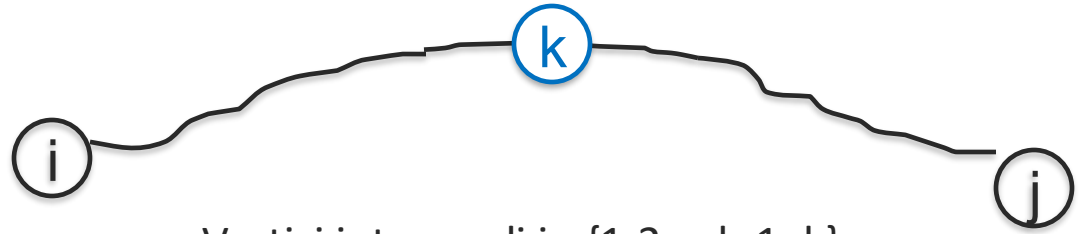
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=0, b=0$

$$\Rightarrow e_{ij}^{k,0,0} =$$



Vertici intermedi in $\{1,2,\dots,k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,0,0}$$

$$(f = 0 \wedge f' = 0) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

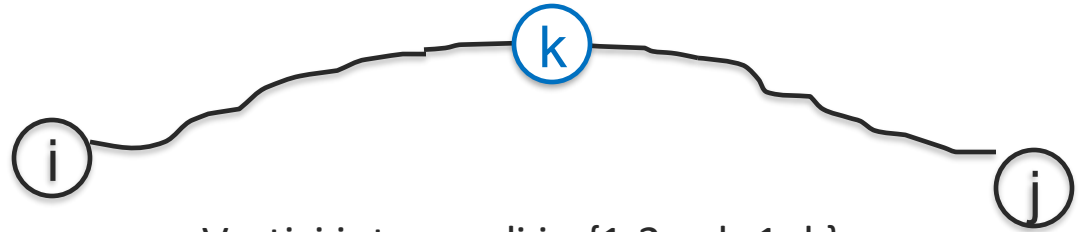
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=0, b=0$

$$\Rightarrow e_{ij}^{k,0,0} = e_{ik}^{k-1,0,0} \wedge e_{kj}^{k-1,0,0}$$



$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,0,0}$$

$$(f = 0 \wedge f' = 0) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

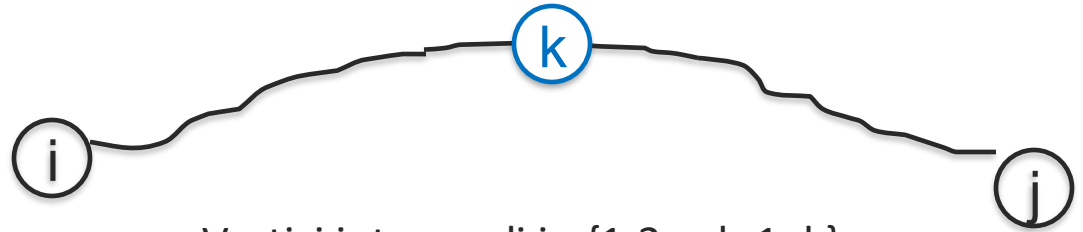
$k > 0, r, b \in \{0, 1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, \underline{r}=0, \underline{b}=0$

$$\Rightarrow e_{ij}^{k,0,0} = e_{ik}^{k-1,0,0} \wedge e_{kj}^{k-1,0,0}$$



Vertici intermedi in $\{1, 2, \dots, k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,0,0}$$

$$(f = 0 \wedge f' = 0) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

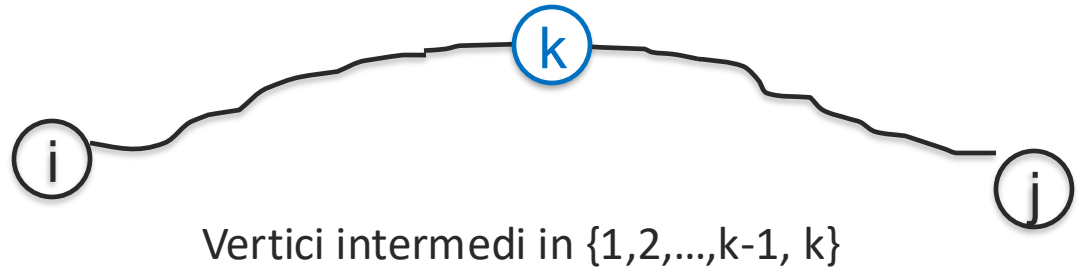
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$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=0, b=0$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ik}^{k-1,r,b} \wedge e_{kj}^{k-1,r,b}$$



Eq. Ricorrenza – Passo ricorsivo

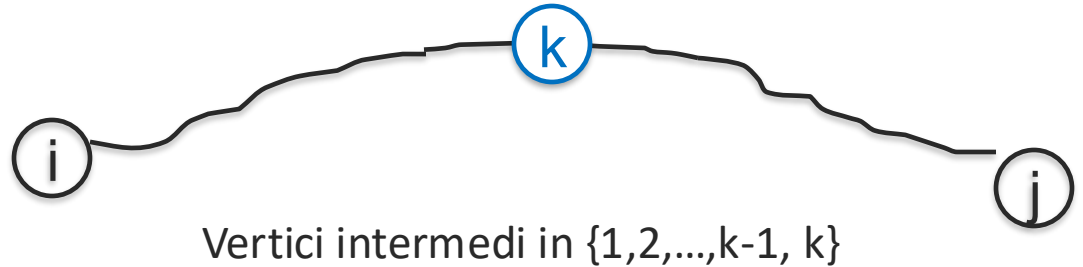
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$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=1, b=0$$

$$\Rightarrow e_{ij}^{k,1,0} =$$



$$\Rightarrow \exists P_{ij}^{k,1,0}$$

Eq. Ricorrenza – Passo ricorsivo

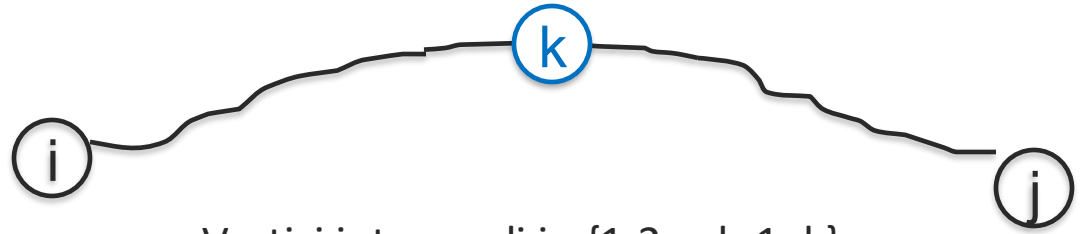
$k > 0, r, b \in \{0, 1\}$

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$k \in P_{ij}^{k,r,b}, r=1, b=0$

$\Rightarrow e_{ij}^{k,1,0} =$



$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,1,0}$$

Eq. Ricorrenza – Passo ricorsivo

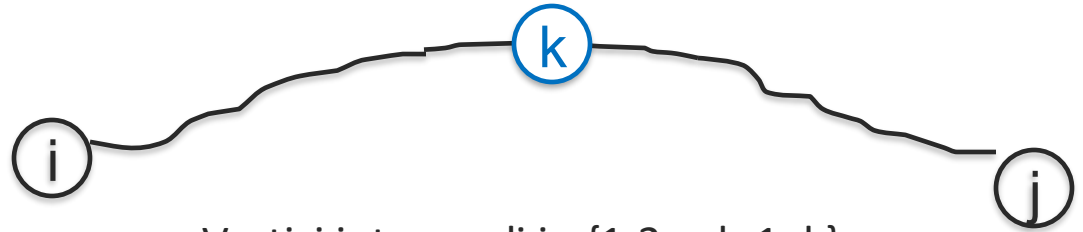
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$k \in P_{ij}^{k,r,b}, r=1, b=0$

$\Rightarrow e_{ij}^{k,1,0} =$



Vertici intermedi in $\{1,2,\dots,k-1, k\}$

$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,1,0}$$

$$(f = 1 \vee f' = 1) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

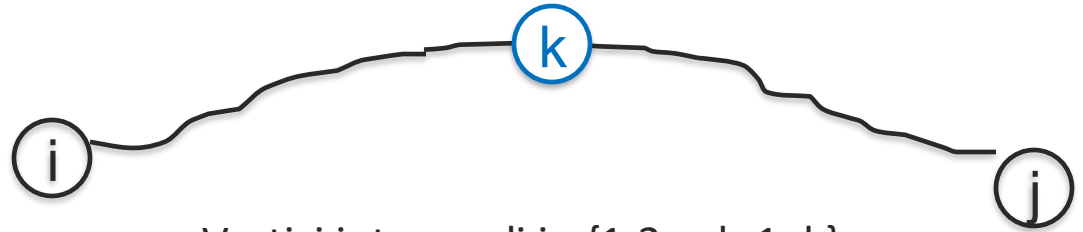
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=1, b=0$

$$\Rightarrow e_{ij}^{k,1,0} =$$



Vertici intermedi in $\{1,2,\dots,k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,0}$$

$$(f = 1 \vee f' = 1) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

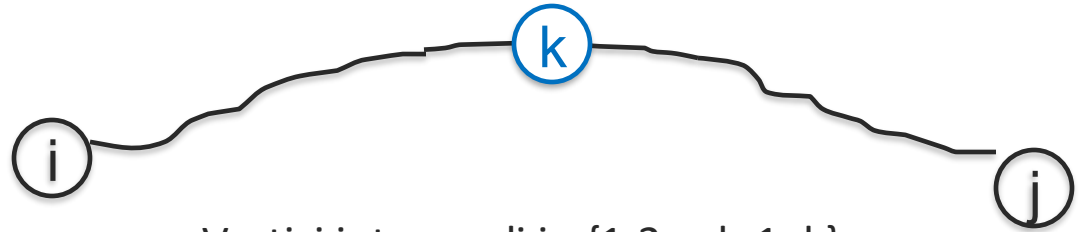
$k > 0, r, b \in \{0, 1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=1, b=0$

$$\Rightarrow e_{ij}^{k,1,0} = \vee \{e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = 1 \vee f' = 1) \wedge (g = 0 \wedge g' = 0)\}$$



Vertici intermedi in $\{1, 2, \dots, k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,0}$$

$$(f = 1 \vee f' = 1) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

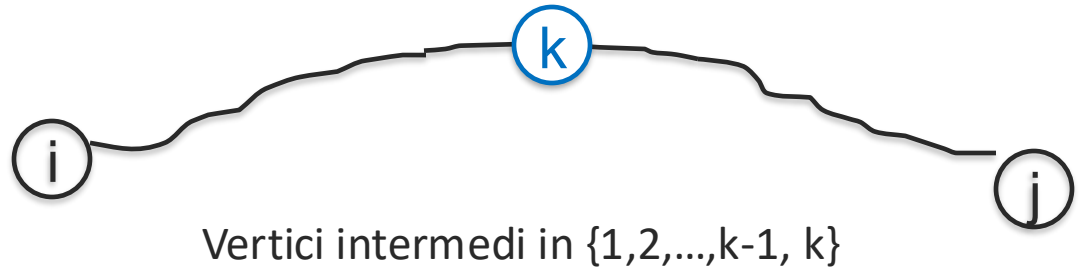
$$k > 0, r, b \in \{0, 1\}$$

$$k \notin P_{ij}^{k,r,b}$$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, \underline{r}=1, \underline{b}=0$$

$$\Rightarrow e_{ij}^{k,1,0} = \vee \{ e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = \underline{1} \vee f' = \underline{1}) \wedge (g = \underline{0} \wedge g' = \underline{0}) \}$$



$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,0}$$

$$(f = 1 \vee f' = 1) \wedge (g = 0 \wedge g' = 0)$$

Eq. Ricorrenza – Passo ricorsivo

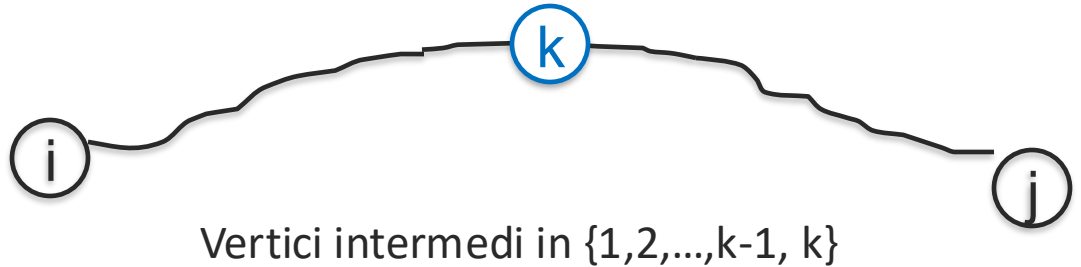
$$k > 0, r, b \in \{0,1\}$$

$$k \notin P_{ij}^{k,r,b}$$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=1, b=0$$

$$\Rightarrow e_{ij}^{k,r,b} = \vee \{e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = r \vee f' = r) \wedge (g = b \wedge g' = b)\}$$



Eq. Ricorrenza – Passo ricorsivo

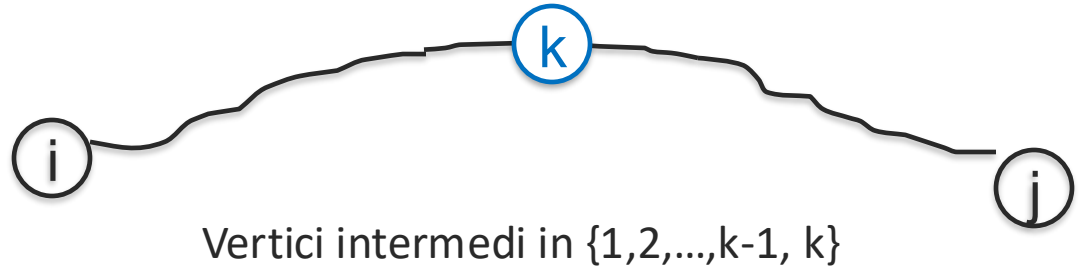
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=1, b=0$

$$\Rightarrow e_{ij}^{k,r,b} = \vee \{e_{ik}^{k-1,f,b} \wedge e_{kj}^{k-1,f',b} \mid (f = r \vee f' = r)\}$$





Eq. Ricorrenza – Passo ricorsivo

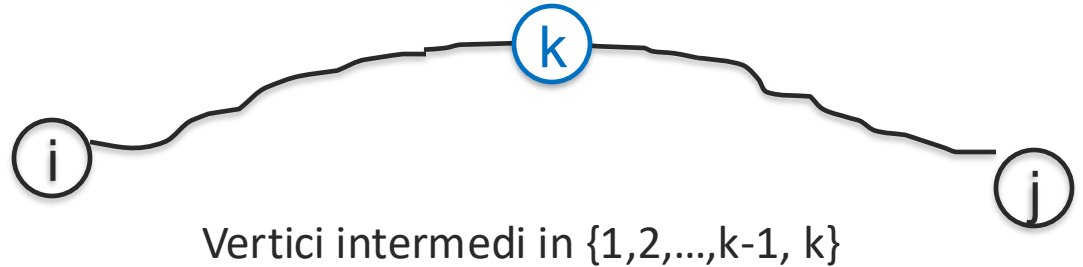
$$k > 0, r, b \in \{0,1\}$$

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Eq. Ricorrenza – Passo ricorsivo

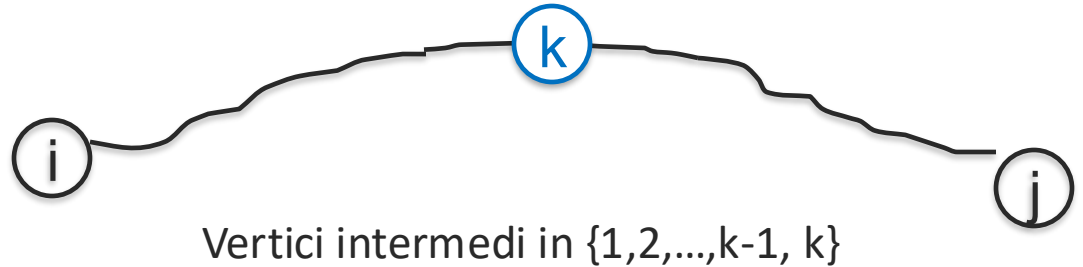
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=0, b=1$

$$\Rightarrow e_{ij}^{k,r,b} = \vee \{e_{ik}^{k-1,r,g} \wedge e_{kj}^{k-1,r,g'} \mid (g = b \vee g' = b)\}$$



Eq. Ricorrenza – Passo ricorsivo

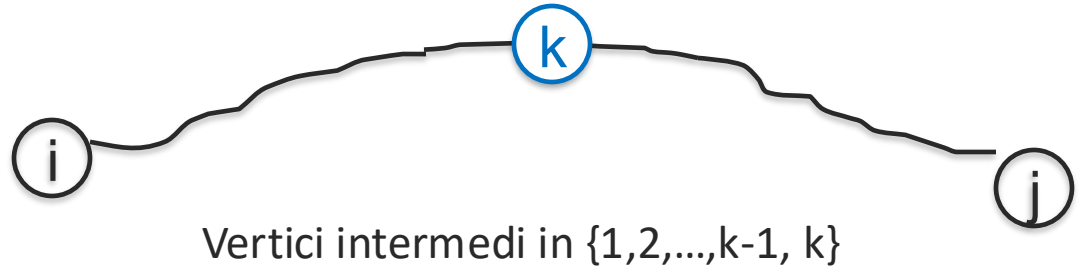
$$k > 0, r, b \in \{0, 1\}$$

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$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=1, b=1$$

$$\Rightarrow e_{ij}^{k,1,1} =$$



$$\Rightarrow \exists P_{ij}^{k,1,1}$$

Eq. Ricorrenza – Passo ricorsivo

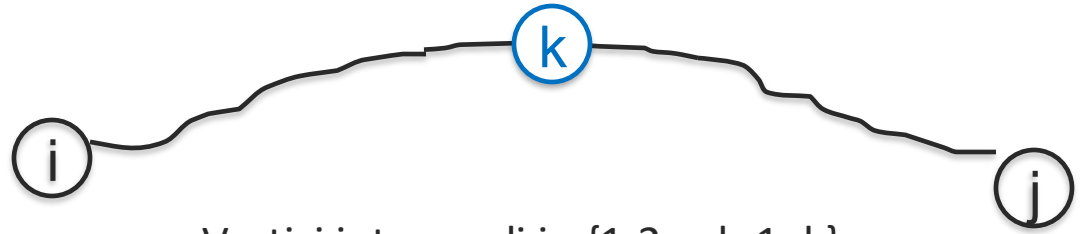
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$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,1,1}$$

Eq. Ricorrenza – Passo ricorsivo

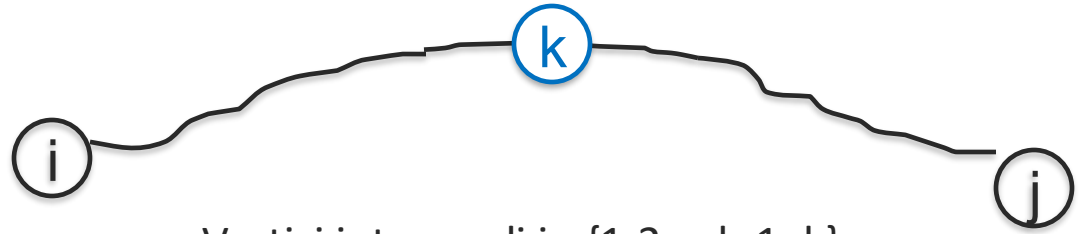
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$k \in P_{ij}^{k,r,b}, r=1, b=1$

$\Rightarrow e_{ij}^{k,1,1} =$



$$\exists P_{ik}^{k-1,f,g} \wedge \exists P_{kj}^{k-1,f',g'} \Rightarrow \exists P_{ij}^{k,1,1}$$

$$(f = 1 \vee f' = 1) \wedge (g = 1 \vee g' = 1)$$

Eq. Ricorrenza – Passo ricorsivo

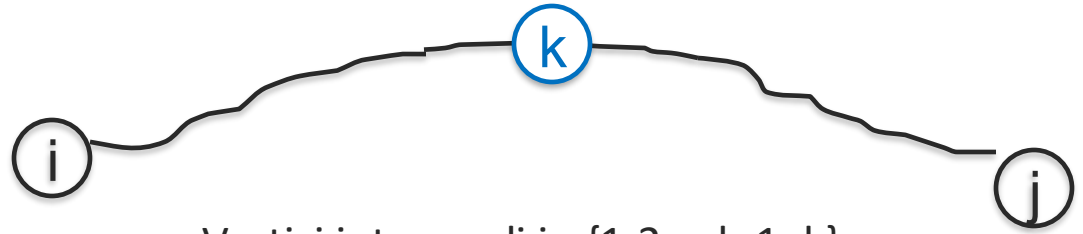
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=1, b=1$

$$\Rightarrow e_{ij}^{k,1,1} =$$



Vertici intermedi in $\{1,2,\dots,k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,1}$$

$$(f = 1 \vee f' = 1) \wedge (g = 1 \vee g' = 1)$$



Eq. Ricorrenza – Passo ricorsivo

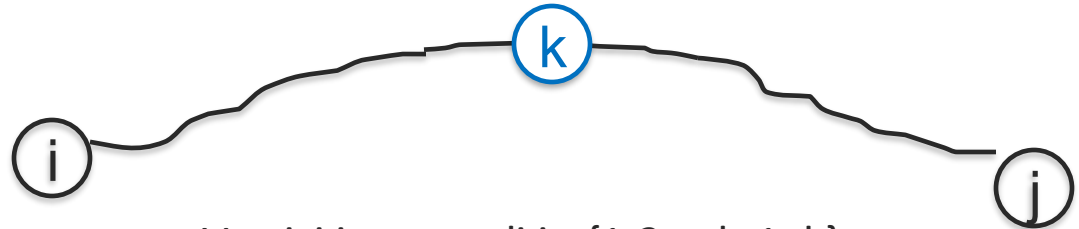
$k > 0, r, b \in \{0,1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, r=1, b=1$

$$\Rightarrow e_{ij}^{k,1,1} = \vee \{ e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = 1 \vee f' = 1) \wedge (g = 1 \vee g' = 1) \}$$



$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,1}$$

$$(f = 1 \vee f' = 1) \wedge (g = 1 \vee g' = 1)$$



Eq. Ricorrenza – Passo ricorsivo

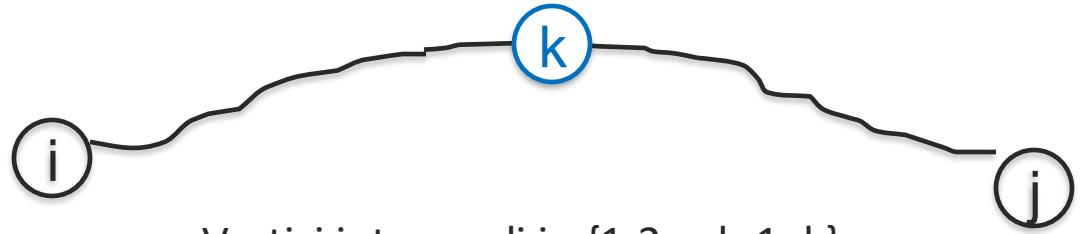
$k > 0, r, b \in \{0, 1\}$

$k \notin P_{ij}^{k,r,b}$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$k \in P_{ij}^{k,r,b}, \underline{r}=1, \underline{b}=1$

$$\Rightarrow e_{ij}^{k,\underline{1},\underline{1}} = \vee \{ e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = \underline{1} \vee f' = \underline{1}) \wedge (g = \underline{1} \vee g' = \underline{1}) \}$$



Vertici intermedi in $\{1, 2, \dots, k-1, k\}$

$$e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \Rightarrow e_{ij}^{k,1,1}$$

$$(f = 1 \vee f' = 1) \wedge (g = 1 \vee g' = 1)$$

Eq. Ricorrenza – Passo ricorsivo

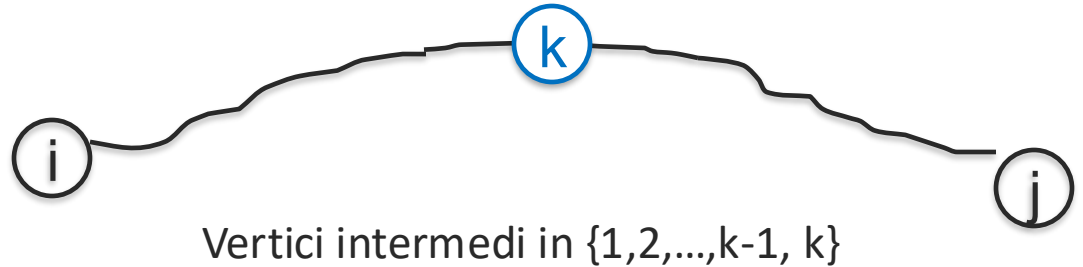
$$k > 0, r, b \in \{0,1\}$$

$$k \notin P_{ij}^{k,r,b}$$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$

$$k \in P_{ij}^{k,r,b}, r=1, b=1$$

$$\Rightarrow e_{ij}^{k,r,b} = \vee \{e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = r \vee f' = r) \wedge (g = b \vee g' = b)\}$$

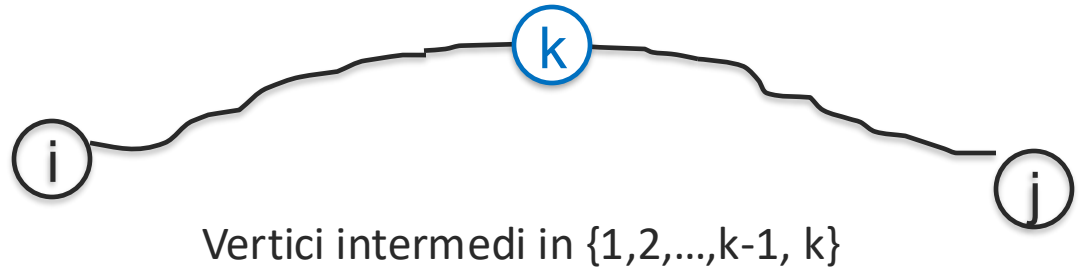


Eq. Ricorrenza – Passo ricorsivo

$$k > 0, r, b \in \{0, 1\}$$

$$k \notin P_{ij}^{k,r,b}$$

$$\Rightarrow e_{ij}^{k,r,b} = e_{ij}^{k-1,r,b}$$



$$k \in P_{ij}^{k,r,b}, r=1, b=1$$

$$\Rightarrow e_{ij}^{k,r,b} = \vee \{e_{ik}^{k-1,f,g} \wedge e_{kj}^{k-1,f',g'} \mid (f = r \vee f' = r) \wedge (g = b \vee g' = b)\}$$

Ricordarsi che per, ogni $(r, b) \in \{0, 1\}^2$,
 $e_{ij}^{k,r,b}$ è da calcolare come la disgiunzione logica tra i due casi

$$k \notin P_{ij}^{k,r,b} \text{ e } k \in P_{ij}^{k,r,b}$$